

# SOUJANYA CHAVAN

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## CAREER SUMMARY

GenAI and ML engineer ready to ship from day one. Built **LLM pipelines, RAG systems, and transformer-based NLP models** (86% accuracy on 50K+ tweets/month) across production and research environments. Trained **GLM models at 76% directional accuracy across 50+ countries**, cut analysis time by **50%**, and drove **17% product adoption uplift** via A/B experimentation. Fluent in LangChain, PyTorch, and AWS Bedrock. Thrives in fast-moving environments where models go from notebook to production in days.

## TECHNICAL SKILLS

**Generative AI & LLM Engineering:** LLM app development, prompt engineering, embeddings, RAG pipelines, vector search; conversational memory, agent workflows, hallucination mitigation.

**Machine Learning & Transformers:** PyTorch, TensorFlow; RNN, LSTM, transformers; supervised/unsupervised learning; AUC, F1, RMSE evaluation.

**NLP & Data Pipelines:** Python, SQL, Pandas, NumPy, Spark; text preprocessing, tokenization, TF-IDF, chunking; structured and unstructured pipelines.

**MLOps & Deployment:** LangChain, Streamlit, Docker, CI/CD, SQLAlchemy; AWS (S3, Athena, SageMaker, Bedrock); FAISS, Chroma, Pinecone; model monitoring.

## PROFESSIONAL EXPERIENCE

**AI Engineer Intern — Borealis Group Analytics — Blythewood, SC** July 2025–September 2025

- Built **ML pipelines** on multi-country economic datasets enabling predictive and **LLM-driven analytics**.
- Developed **regression/GLM models achieving 76% directional accuracy** across 50+ countries.
- Engineered **30+ years of macroeconomic features** (GDP, FDI, FX, market cap) for model training and inference.
- Integrated structured model outputs into **LLM workflows (AWS Bedrock)** for automated analytical summaries.
- Validated **model and LLM outputs** against benchmark equity-factor datasets to ensure reliability.

**Machine Learning Research Assistant — USC Information Sciences — Los Angeles, CA** October 2024–January 2025

- Built scalable **NLP datasets processing 50K+ tweets/month** for election influence analysis.
- Trained **transformer models**: ELECTRA 86% influence accuracy, RoBERTa 84% ideology accuracy.
- Applied **LDA topic modeling** to extract large-scale political discourse themes.

**Machine Learning Engineer — Motleyscape — India** August 2023–June 2024

- Built **ML pipelines** processing AR/VR telemetry and interaction datasets for behavioral analysis.
- Drove **17% improvement in AR feature adoption** via A/B experimentation and statistical validation.
- Developed **NLP pipelines on 10K+ feedback records** to surface UX insights and guide product decisions.
- Prototyped **RNN/LSTM sequence models** for user interaction pattern prediction in XR environments.
- Automated **feature engineering workflows reducing manual analysis by 50%** across the ML lifecycle.
- Designed **AI-powered dashboards** translating model outputs into actionable product insights.

## EDUCATION

**University of Southern California, CA — Master of Science in Analytics** August 2024–Present

**APSIT Mumbai University, India — Bachelor of Engineering in Computer Science** May 2020–May 2024

## ACADEMIC PROJECTS

**End-to-End Conversational Q&A Chatbot with Message History** June 2024–August 2024

- Launched a **stateful LLM chatbot** with multi-session memory and persistent context.
- Implemented **prompt templates, token trimming, and context-window management** for efficient inference.
- Improved response relevance across **10+ multi-turn queries per session**.

**RAG-Based Document Q&A System** September 2022–December 2022

- Built end-to-end **RAG pipeline** indexing chunked documents for grounded, hallucination-resistant QA.
- Configured **vector retrieval with history-aware query reformulation** for contextually accurate answers.
- Constrained LLM responses to retrieved context, **measurably reducing hallucinations**.

**Chat with SQL Database using LangChain** August 2025–October 2025

- Built a **GenAI SQL copilot** converting natural language into executable SQL queries.
- Enabled **schema-aware querying** across SQLite and MySQL with dynamic table context injection.
- Reduced manual SQL dependency via **conversational analytics workflows**.

**LSTM / RNN Next-Word Prediction & Sentiment Modeling** August 2025–December 2025

- Formulated **LSTM-based next-word prediction models** for sequence generation tasks.
- Executed **IMDB sentiment classification** with padded sequences and threshold tuning.
- Deployed **real-time inference demo** showcasing tokenization and confidence scoring.